

VS Code for Data Science

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What is VS Code?

- **Visual Studio Code** is a code editor / integrated development environment (IDE)
 - Lightweight, adaptable for almost any language
 - Can be geared for specific languages using official or community-built **extensions**
 - Combines code and text editing, file system navigation, terminal window, debugging, and AI integration through **GitHub Copilot**
- Not to be confused with **Visual Studio**, a full IDE with compiler meant for C#, C++, and other specific languages



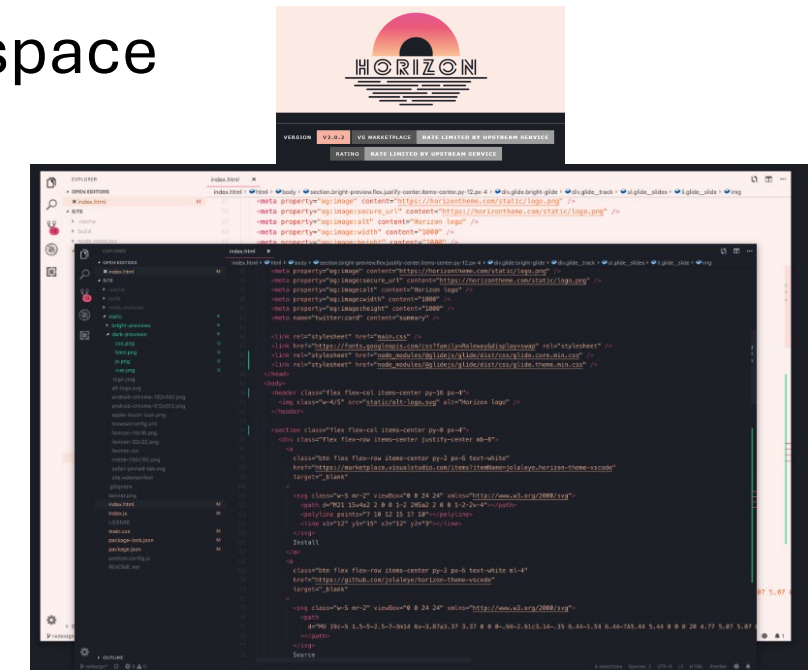
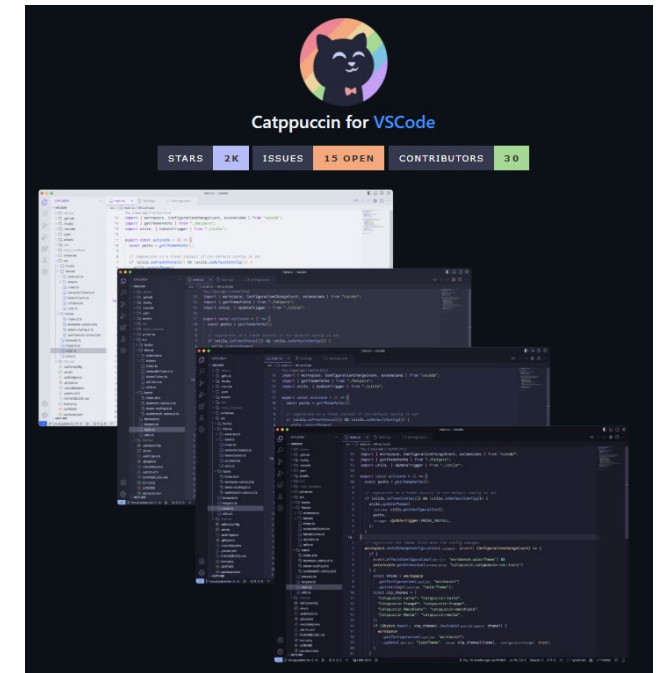
**Visual Studio
Code**



Visual Studio

Why should you use VS Code?

- VS Code is lightweight and highly customizable
 - Can run on almost any machine
 - You choose the relevant tools for your project
 - More aesthetic than RStudio
- VS Code combines many useful functions into one space
 - Lowers the learning curve for useful tools
 - Makes good practices easier to incorporate
 - Incentivizes designing your own workflow
 - Provides you with translatable skillsets
- **Most importantly – VS Code saves you time!!**



Before VS Code ☹️

- You might have to:
 - Create and move a lot of files and folders – **Drag and drop in File Explorer**
 - Access and move data between multiple servers – **USB sticks and Dropbox**
 - Write your own scripts in R, Bash, or Python – **Switch between Rstudio, Powershell, and Spyder**
 - Debug those scripts when they don't work – **Copy and paste code into Google / ChatGPT**
 - Use version control software for reproducibility – **Upload everything to GitHub once a year**
 - Open and preview many different image formats – **Random software and ImageJ windows everywhere**
 - Open, import, and edit many different tabular data formats – **Excel turns everything into dates**
 - Install different versions of specialized packages – **Reinstalling Java for the tenth time**



After VS Code 😊



- You might have to:
 - Create and move a lot of files and folders – **Integrated terminal and folder navigator!**
 - Access and move data between multiple servers – **SSH, SCP, and Globus!**
 - Write your own scripts in R, Bash, or Python – **Different specialized official extensions in one window!**
 - Debug those scripts when they don't work – **Integrated GitHub Copilot – no more pasting!**
 - Use version control software for reproducibility – **Full Git integration – easy pushes and pulls!**
 - Open and preview many different image formats – **Specialized extensions and tabs!**
 - Open, import, and edit many different tabular data formats – **Unix, Rainbow CSV, and Data Wrangler!**
 - Install different versions of specialized packages – **Works easily with conda environments and containers!**

R in VS Code

The screenshot displays the VS Code interface with an R project. The left sidebar shows the workspace with a data table and a list of models. The main editor shows the R script `data_analysis.R` with the following code:

```
library(data.table)

set.seed(123)
n <- 1000
dt <- data.table(id = 1:n)
dt[, x1 := rnorm(.N, mean = 0, sd = 2)]
dt[, x2 := runif(.N, min = -1, max = 1)]
dt[, y := 2 * x1 + x2 + 0.5 * rnorm(.N)]

model <- lm(y ~ x1 + x2, data = dt)

summary(model)
plot(model)
```

The right sidebar shows the R console output, including the R version (4.1.3), the data table creation, the linear model fit, and the summary statistics:

```
Call:
lm(formula = y ~ x1 + x2, data = dt)

Residuals:
    Min       1Q   Median       3Q      Max
-1.54376 -0.31646 -0.01093  0.34316  1.61131

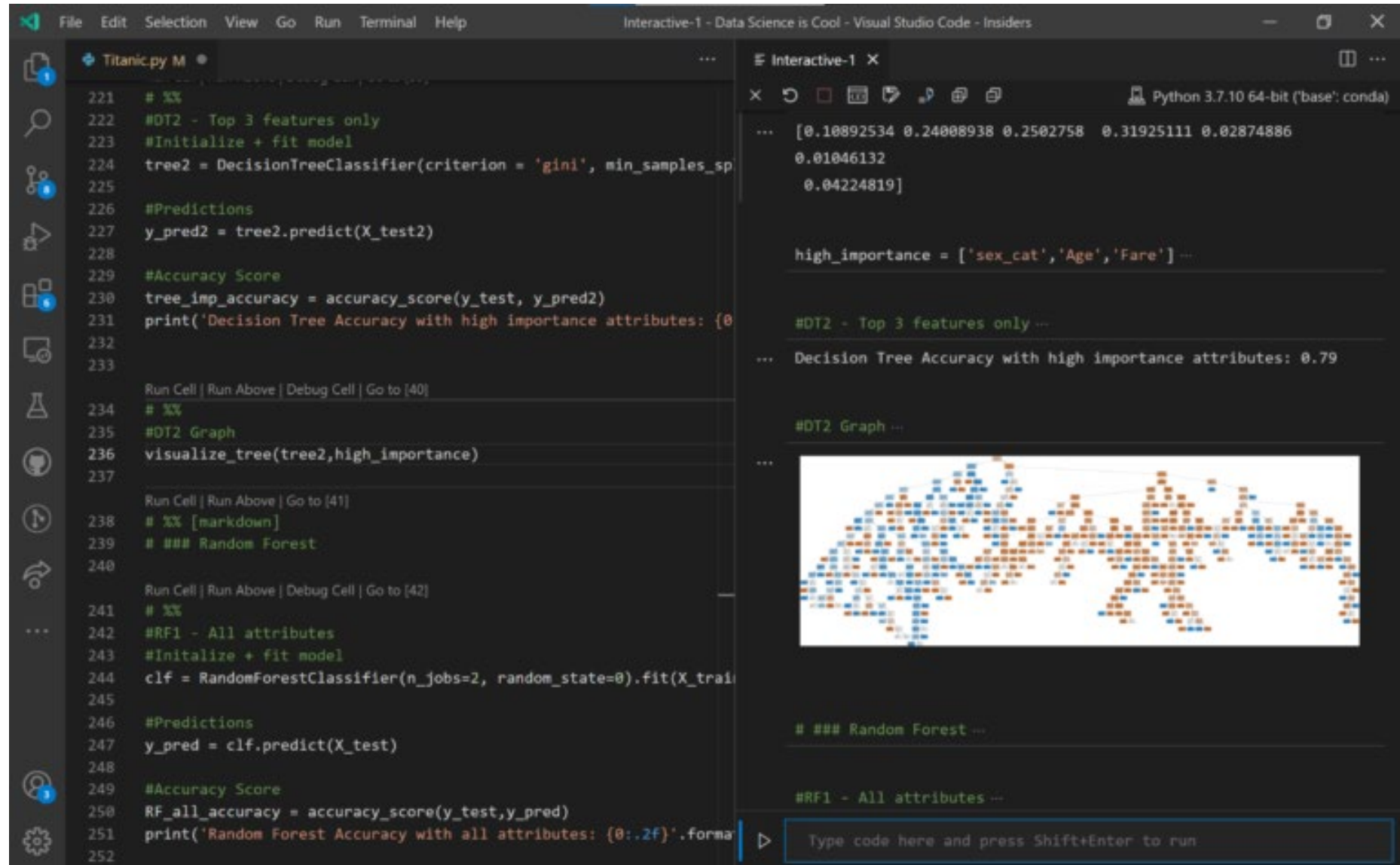
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0007977  0.0155577  -0.051   0.959
x1           1.9855040  0.0078490 252.963 <2e-16 ***
x2           1.0224384  0.0266384  38.382 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4919 on 997 degrees of freedom
Multiple R-squared:  0.9851,    Adjusted R-squared:  0.9851
F-statistic: 3.298e+04 on 2 and 997 DF,  p-value: < 2.2e-16
```

The bottom plot shows the Residuals vs Leverage plot, which is a scatter plot of standardized residuals against leverage. The plot includes a red horizontal line at zero and a dashed red line for Cook's distance. The x-axis is labeled 'Leverage' and 'lm(y ~ x1 + x2)', and the y-axis is labeled 'Standardized residuals'. The plot shows a dense cluster of points with a few outliers labeled with their IDs (e.g., 867, 2650, 4390).

The bottom status bar shows the R version (4.1.3), the number of lines (25159), the current line (Ln 14, Col 1), the number of spaces (2), the encoding (UTF-8), and the language (R).

Python in VS Code



The screenshot displays the Visual Studio Code interface with a Python file named 'Titanic.py' open. The code is divided into two sections: a Decision Tree model and a Random Forest model. The Decision Tree section includes comments for feature selection, model initialization, prediction, and accuracy calculation. The Random Forest section follows a similar structure. The interactive console on the right shows the output of the code, including a list of feature importance values, the top 3 features, and a decision tree visualization. The Random Forest section is partially visible at the bottom.

```
221 # %%
222 #DT2 - Top 3 features only
223 #Initialize + fit model
224 tree2 = DecisionTreeClassifier(criterion = 'gini', min_samples_sp
225
226 #Predictions
227 y_pred2 = tree2.predict(X_test2)
228
229 #Accuracy Score
230 tree_imp_accuracy = accuracy_score(y_test, y_pred2)
231 print('Decision Tree Accuracy with high importance attributes: {0
232
233
234 Run Cell | Run Above | Debug Cell | Go to [40]
235 # %%
236 #DT2 Graph
237 visualize_tree(tree2,high_importance)
238
239 Run Cell | Run Above | Go to [41]
240 # %% [markdown]
241 # ### Random Forest
242
243 Run Cell | Run Above | Debug Cell | Go to [42]
244 # %%
245 #RF1 - All attributes
246 #Initialze + fit model
247 clf = RandomForestClassifier(n_jobs=2, random_state=0).fit(X_train
248
249 #Predictions
250 y_pred = clf.predict(X_test)
251
252 #Accuracy Score
253 RF_all_accuracy = accuracy_score(y_test,y_pred)
254 print('Random Forest Accuracy with all attributes: {0:.2f}'.forma
255
```

Interactive-1 X
Python 3.7.10 64-bit ('base': conda)

[0.10892534 0.24008938 0.2502758 0.31925111 0.02874886
0.01046132
0.04224819]

high_importance = ['sex_cat', 'Age', 'Fare'] ...

#DT2 - Top 3 features only ...

... Decision Tree Accuracy with high importance attributes: 0.79

#DT2 Graph ...

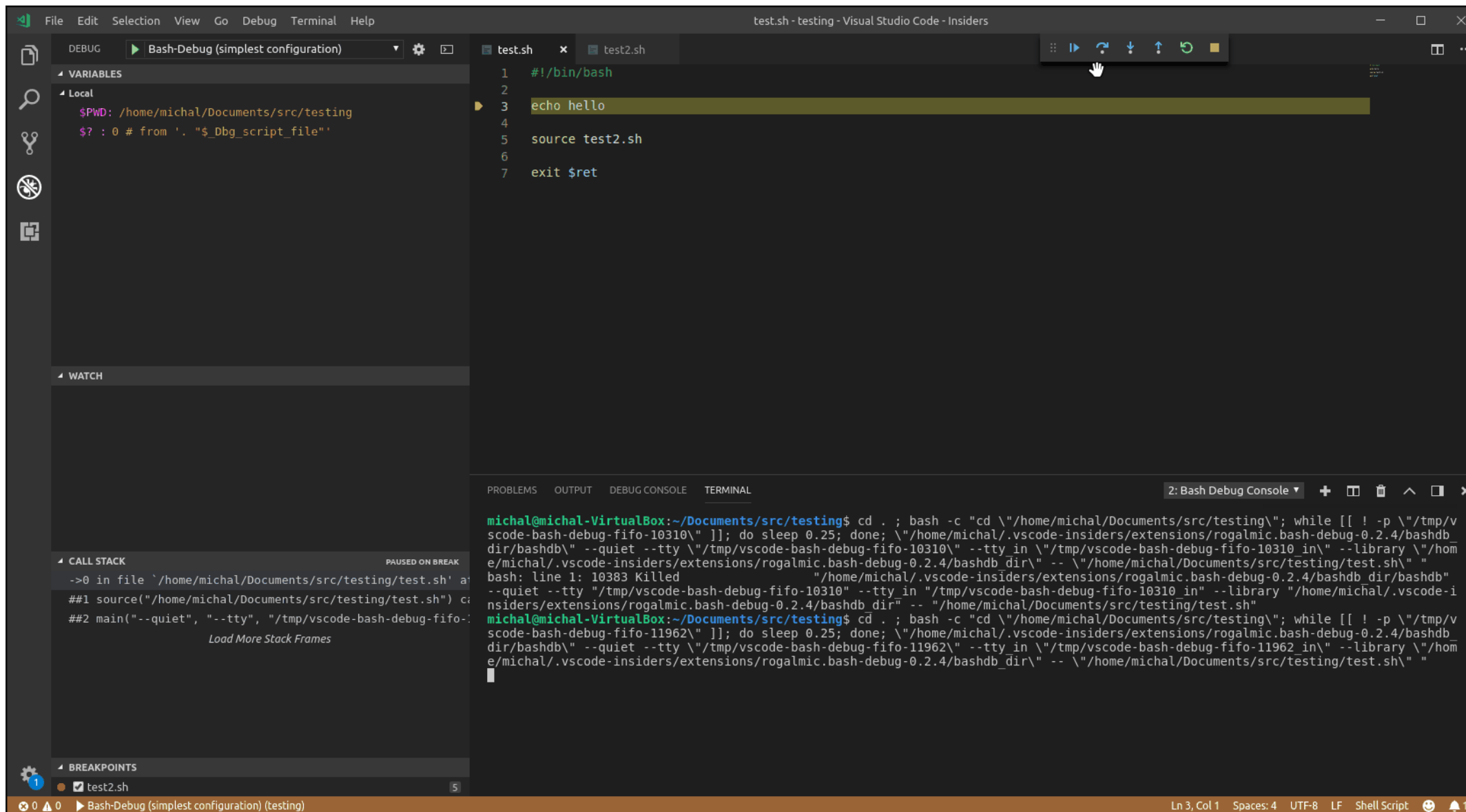
...

Random Forest ...

#RF1 - All attributes ...

Type code here and press Shift+Enter to run

Bash scripting in VS Code



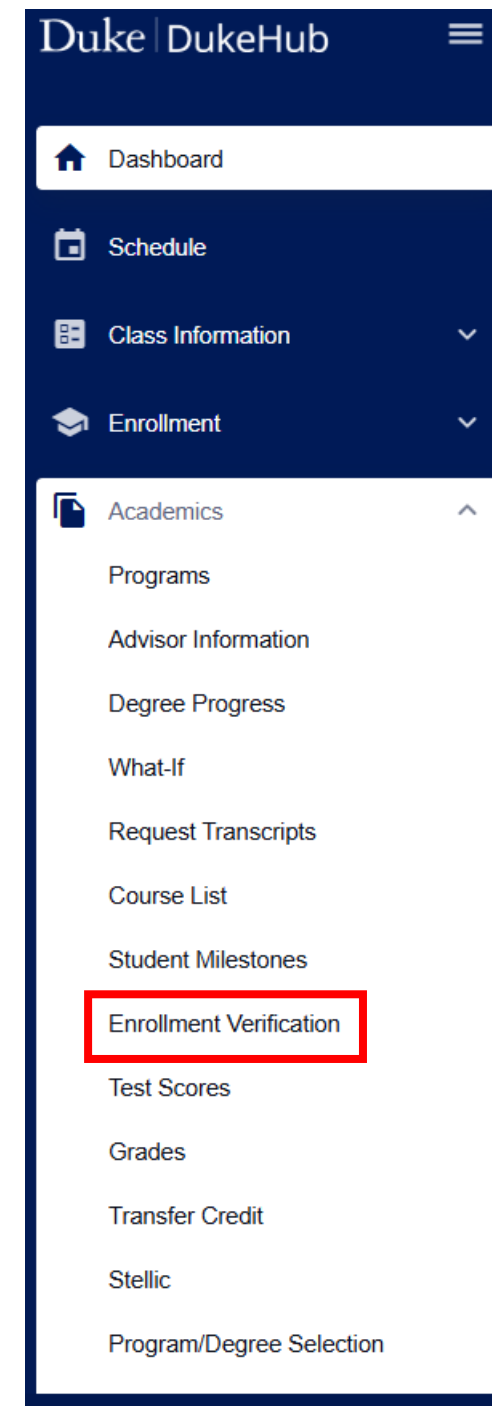
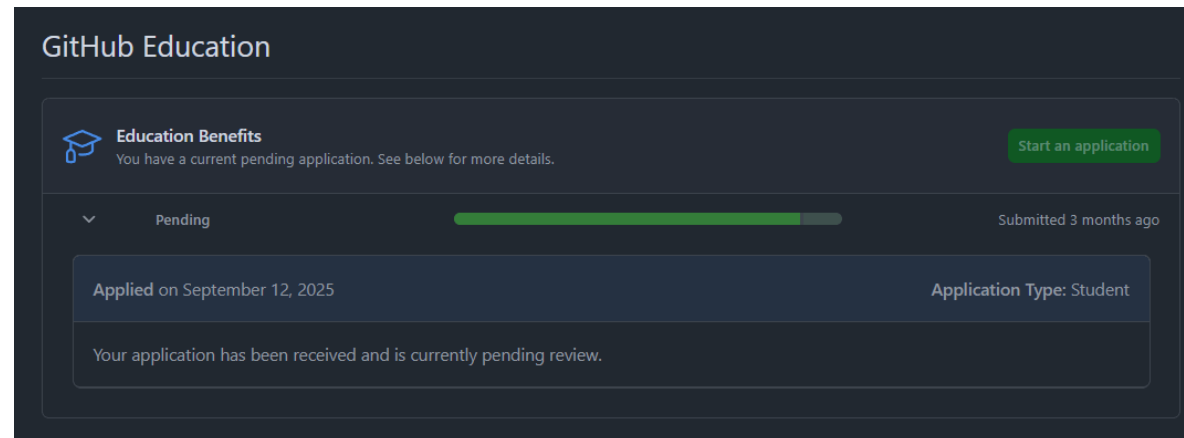
Getting started

Full setup tutorial: https://github.com/magwenelab/CodeStarterKit/blob/main/vscode_setup.md

- Install VS Code: <https://code.visualstudio.com/Download>
- Install recommended basic extensions
 - Remote Development
 - **Remote – SSH: Editing Configuration Files**
 - **Remote Explorer**
 - **Remote – Tunnels**
 - For WSL
 - **Remote Repositories**
 - Copy/download files directly from repositories without cloning!
 - Language-specific
 - **Jupyter (+ cell tags, renderers, slide show)**
 - **Python (+ Pylance, Debugger)**
 - **R (+ Debugger)**
 - **Snakemake Language**
 - Quality of Life
 - **Rainbow CSV**
 - **Edit CSV**
 - Excel-like sheets interface
 - **Data Wrangler**
 - Powerful python-based data frame manipulation interface!
 - **SVG Preview**
 - **vscode-pdf**
 - **filesize**
 - Shows file size info + other metadata interface
 - Copilot
 - **GitHub Copilot**
 - **GitHub Copilot Chat**

Getting GitHub Copilot

- GitHub Copilot is free through **GitHub Education**
 - Requires proof of student status
- DukeHub -> Academics -> Enrollment Verification
 - “submit” and either screenshot or convert the pdf to an image
 - Make sure to get the “as of” date and “Exp Comp Dt”
- Start a GitHub Education Student application and upload the image
- Wait a 3-5 days
- Link to VS Code



- **Globus** offers a fast, secure option for large file transfers between Duke servers (e.g. to/from the DCC)
 - <https://github.com/magwenelab/CodeStarterKit/blob/main/globus.md>
- Tired of typing in your password every time you ssh? Generate a **ssh key pair** and upload the public key to Duke OIT
 - Make sure to leave it blank when it asks for a passphrase during key generation
 - <https://adhilton.pratt.duke.edu/ssh-key-pair-setup-upload>

Manage Your Public SSH Keys

Your public ssh keys can be used by some applications or servers for authentication.

Public SSH Keys Registered + See More about SSH keys

You have 3 keys registered.

- Adding the ssh key to your ssh config for VS Code can be complicated
 - Follow this tutorial or reach out if you need help
 - https://github.com/magwenelab/CodeStarterKit/blob/main/ssh_keys.md